Blockchain and Federated Learning for Collaborative Intrusion Detection in Vehicular Edge Computing

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***Abstract*—The vehicular networks constructed by interconnected vehicles and transportation infrastructure are vulnerable to cyber-intrusions due to the expanded use of software and the introduction of wireless interfaces. Intrusion detection systems (IDSs) can be customized efficiently in response to this increased attack surface. There has been significant progress in detecting malicious attack traffic using machine learning approaches. However, existing IDSs require network devices with powerful computing capabilities to continuously train and update complex network models, which reduces the efficiency and defense capability of intrusion detection systems due to limited resources and untimely model updates. This work proposes a cooperative intrusion detection mechanism that offloads the training model to distributed edge devices (e.g., connected vehicles and roadside units (RSUs). Distributed federated-based approach reduces resource utilization of the central server while assuring security and privacy. To ensure the security of the aggregation model, blockchain is used for the storage and sharing of the training models. This work analyzes common attacks and shows that the proposed scheme achieves cooperative privacy-preservation for vehicles while reducing communication overhead and computation cost.**

***Index Terms*—Intrusion detection, federated learning, vehicular networks, blockchain.**

I. Introduction

MODERN autonomous vehicles' attack surfaces are increasing due to expanded software and wireless interfaces, such as vehicles to everything and 5G, that are used to sense and communicate with surrounding environments. This has caused an emerging market for intrusion detection systems (IDS) to mitigate the cybersecurity risks [1]. Vehicular edge computing is an enabling technology for V2X to facilitate the development of smart cities and autonomous driving. The deployment of 5G can greatly improve the efficiency of edge computing and speed up the training of edge intelligence [2]. Secure and privacy-enhanced V2X and 5G links need to be established in the context of collaborative intelligent systems [3], [4].

Deep learning-based anomaly-detection IDSs are being developed to improve the accuracy of attack detection as the main goal of such systems. Traditional machine learning algorithms (e.g., support vector machines (SVM), decision trees, random forests) and a variety of deep neural networks, such as convolutional neural network (CNN), recurrent neural network (RNN), and generative adversarial neural network (GAN) have been widely used in the field [5]. For example, a machine learning-based framework has been developed for detecting Distributed Denial of Service (DDoS) attacks in smart cities, which has been proven to be very effective [6]. Among various machine learning algorithms, deep neural networks seem to have many potentials [7]. A deep learning-based model for replay attack detection has been proposed, which can distinguish normal and abnormal behaviors with relatively high accuracy [8]. However, IDS based on deep learning requires that the devices have powerful computing capacities. Not to mention, the training process can be time-consuming when models have higher complexities on a device. This centralized training task places a computational burden on the central device, which can also be vulnerable to attacks.

Traditional machine learning requires a centralized cloud training server, which is prone to problems of data privacy and leakage. Multi-party data holders are often reluctant to share their training data for data privacy-preserving. However, machine learning-based intrusion detection currently requires training data to train an effective intrusion detection model. Machine learning algorithms without data are challenging to achieve the desired detection results. How to solve the problem of centralized model training and not leaking the native training data, completing the model training is a critical problem.

There have been several approaches proposed, e.g., Konečnỳ *et al.* [9] proposed the blockchain and federated optimization to solve the traditional data training centralization problem, or McMahan *et al.* [10] elicited federated learning and stated that it is a communication-efficient deep network learning [11]. In another work, [12] show federated learning can deliver centralized training tasks to edge nodes. Edge computing has laid the foundation for the realization of federated learning. By increasing computing capabilities in some devices at the

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edge, the computing burden of a centralized data server can be significantly alleviated [13], [14]. While using edge machine learning and sharing the results of the edge learning with traffic infrastructure (base stations or RSUs) via network communication, the computing tasks of the edge facilities are alleviated [15], [16].

Under the premise of legal compliance, efficient machine learning is carried out between multiple participants or multiple computing nodes. As a moving edge computing node, an edge vehicle in a vehicle network has limited computing resources [17], which makes it difficult to update its model for current global vehicles' intrusion detection data. Mobile Crowd-Machine Learning can effectively improve collaboration efficiency [18]. To obtain a new updating model for global training data from different vehicles, federated learning can be used to assist edge vehicles in training and obtaining the global model updates without knowing the data privacy of neighbor vehicles and not revealing their private data. Each edge vehicle traveling to a specific RSU area applies for a global model from the RSU and then uses its intrusion detection data to perform edge model training. The RSU implements the model aggregation and sends it to edge vehicles. Under this federated learning, multiple edge vehicles have completed the training and update of the global intrusion detection model collaboratively by communicating with the RSU without leaking their private data.

Federated learning is an encrypted and distributed machine learning method that can overcome the problem of data isolated island [19], [20]. All parties can build a global model without disclosing the underlying raw data. The traditional federated learning model aggregation task is completed by a centralized cloud server. A central server first collects edge model updates (e.g., weights and gradients) and uses them to implement global model aggregation. RSUs play the role of model aggregation in vehicular networks. After the model aggregation, a key issue is the safe disposal of model storage and sharing. Centralized storage often has some risks of tampering and privacy leakage. To prevent a single RSU's trained model from being attacked and maliciously tampered with, the blockchain can be used to solve its centralization aggregation problem. To improve the scalability of an RSU, multiple RSUs can jointly maintain an intrusion detection blockchain to achieve the purpose of preventing tampering of the aggregate model and complete the secure model storage and sharing task. To better train a high-accuracy network model, this work proposes a trust-based consensus mechanism. Intelligent trust evaluation and management for sensor nodes can effectively improve the reliability of edge computing nodes [21], [22]. Blockchain was initially used as the underlying support technology for Bitcoin and played a massive role in preventing tampering and decentralization. The application of blockchain is not limited to Bitcoin. The essence of the blockchain is a decentralized, tamper-proof distributed ledger. By building a peer-to-peer network, the blockchain can solve the problem of centralized data storage. Besides, the consensus and incentive mechanism embedded in the blockchain can motivate participants to collaboratively train machine learning models with higher accuracy [23]. The overview of centralized and distributed architectures for federated learning is shown in Fig. 1.

In this work, the authors propose a framework that integrates jointly efficient model training and secure model sharing among vehicles enabled by federated learning and blockchain to assure autonomous driving safety. The main contributions are as follows:

- Design of a two-stage IDS by leveraging federated learning through multiple edge vehicles and roadside facilities (e.g., base stations, RSUs) that collaborate together. In the upper aggregation layer, the traditional centralized cloud model aggregation task is replaced by a distributed collaborative RSUs.
- A mask noise model upload algorithm based on secret sharing is designed to ensure the safety of model upload parameters. Combined with mask noise, it effectively ensures the security of user data and edge models.
- Design of a trust calculation and block verification algorithm driven by model quality. By combining this trust evaluation algorithm, this work also discusses the design of a trust-based incentive mechanism and finally achieves a high accuracy rate of the federated learning model training. The distributed model aggregation scheme based on the blockchain ensures the security of the model sharing and storage.

The remainder of this work is organized as follows. Section II presents the related work. Section III provides the system model. Section IV introduces the proposed cooperative intrusion detection architecture. Performance analysis is introduced in Section V. Section VI concludes the work.

II. Related Work

Otoum *et al.* [24] proposed a restricted Boltzmann machine clustered IDS in wireless sensor networks (WSNs) for monitoring known and unknown attacks based on deep learning. Javaid *et al.* [7] proposed an intrusion detection method based on deep learning in which self-taught learning is based on sparse autoencoder and soft-max regression was adopted to develop a network-based IDS. However, they only consider intrusion detection systems under a single system. That is, all intrusion detection data is concentrated on a single system or device. Because it is difficult to gather intrusion detection data only in one place in vehicular networks, multiple devices with different intrusion detection data can collaborate to complete the construction of an intrusion detection system without the centralized aggregation of training data. Nguyen *et al.* [25] proposed a distributed anomaly detection system without human intervention on IoT equipment. By establishing device-type-specific communication, the system can effectively detect compromised IoT devices. However, the final model aggregation uses centralized aggregation without considering the security issues of the centralized aggregation model. Due to its centralized nature, models aggregated by the central server may be tampered with.

Lu *et al.* [19] discussed a collaborative data-sharing architecture based on permissioned blockchain and federated learning. By establishing a decentralized, collaborative, and distributed

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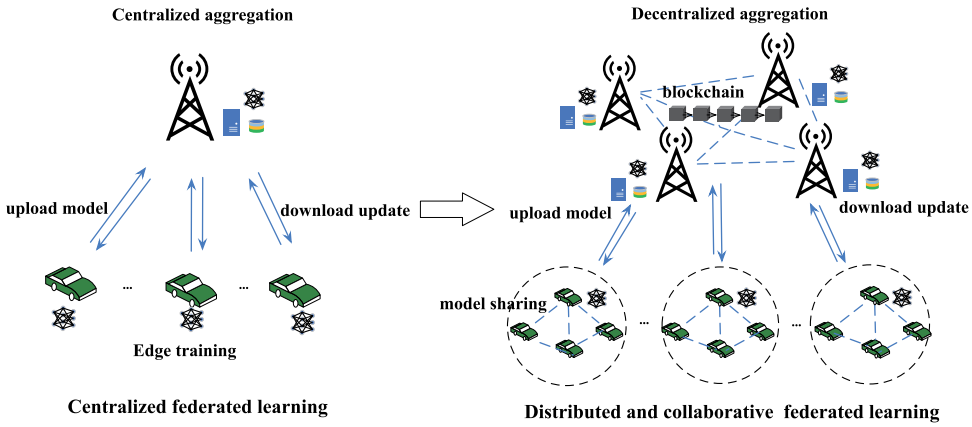


Fig. 1. Overview of centralized and distributed architectures for federated learning.

machine learning system between multiple roadside facilities, the accuracy and safety of the aggregated machine model are ensured. To solve the speculative attacks and privacy leakage caused by data sharing, differential privacy is integrated into federated learning to enhance privacy-preservation. However, this work only considers the establishment of federated learning based on blockchain for RSUs. Though RSUs aggregate the machine learning model and can share the model data to the edge vehicles, the system does not consider the contribution of the edge vehicles to the training data of RSUs. Edge vehicles can also use their edge data to perform edge training so that the edge training results are shared with the RSU by V2X and collaborate to complete the training of the machine learning model.

Wang *et al.* [26] introduced a collaborative intrusion detection network and emphasized data and trust management challenges for IDS. Besides, they discussed the potential of blockchain in building collaborative intrusion systems. However, it did not make specific implementation and scheme design for trust management of cooperative intrusion detection system. Preuveneers *et al.* [27] constructed an anomaly detection model based on blockchain in which a distributed ledger was established by chaining multiple intrusion detection models to achieve the traceability of each intrusion detection model, thereby preventing malicious tampering and malicious poisoning attacks. However, this work uses a private and permissioned blockchain and does not consider the trust issue of aggregation nodes. As a private blockchain, it does not better realize its decentralized characteristics.

Odesile *et al.* [28] proposed an IDS based on multiple mobile agents in which different and distributed edge nodes execute the machine learning process. The mobile gateway device can be used for aggregation purposes. However, this work does not consider the privacy leakage problems faced by different edge nodes when sharing sub-models with the gateway. Due to the lack of a corresponding trust system between various networks represented by different gateways, it is also difficult to achieve secure information sharing.

In this work, the authors propose a cooperative IDS based on federated learning and blockchain. Blockchain is particularly used to ensure privacy-preservation for vehicles and high accuracy model aggregation training for RSUs in vehicular networks.

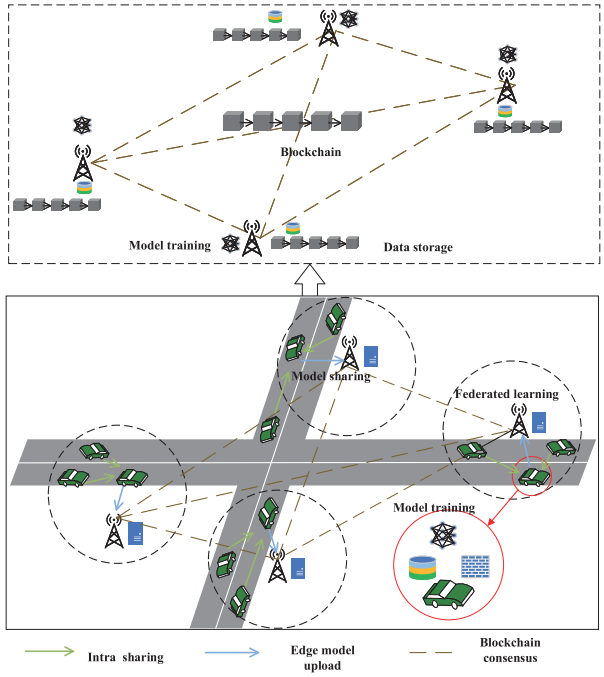


Fig. 2. The proposed systems model.

III. System Model

This section discusses the system model that combines both federated learning and blockchain for efficient model training and secure sharing, as shown in Fig. 2.

A. Cooperative Intrusion Detection Model

The system model is built on a two-stage architecture. In the first stage, vehicles traveling towards an RSU area in a certain period can obtain a pre-trained intrusion detection network model from RSU, and itself acts as an IDS and using new traffic data to train and update further based on the existing model. In the second stage, each vehicle uploads its model trained to the RSU. The RSU can aggregate the network models of multiple vehicles to train and obtain the final global network model. Multiple RSUs perform the above operations at the same

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time, jointly maintaining a blockchain, storing new aggregated machine learning models in the blockchain to implement the periodical update and anti-tampering of the machine model, ensuring the freshness and security of the model in time. Some of the role components involved in the model are as follows:

- *Roadside units:* RSUs as distributed nodes can serve as mining nodes. Multiple RSUs jointly maintain an intrusion detection model blockchain (i.e., IDM-chain). They collect edge machine learning trained models submitted by edge vehicles in their area to train the new aggregation model. Multiple RSUs compete to verify the aggregation model training transactions and the winner as the miner writes them into the blockchain with a distributed consensus mechanism.
- *Vehicles:* As a dynamic and movable edge computing device, the vehicle has its requirements for intrusion detection. It is equipped with a certain sensor device outside and is equipped with an ECU and CAN bus. Due to its dynamic nature, each moving vehicle has different intrusion detection data. A vehicle driving into a certain RSU area can send model requests to its neighboring RSU to obtain the intrusion detection model in this area. After obtaining the intrusion detection model shared by the RSU, the vehicle can train and update the model based on its intrusion detection data, then upload the updated model to the RSU. The RSU will train to aggregate a new intrusion detection model. There are two types of vehicles: one is ordinary edge vehicles used for distributed edge training based on their intrusion detection data, and the other is the representative vehicle of the group elected by RSU. They are responsible for collecting edge vehicles model data, and then upload the pre-training results of the edge aggregation model to RSU.
- *Certification authority:* The certification authority is used to provide some cipher suites for securing data transmission in the network communication. The centralized certification authority is mainly to complete the registration of vehicles and roadside units in the internet of vehicles and the distribution of secret keys and certificates. The issued certificates are not only used for SSL communication but also for message signing.
- *Decentralized blockchain:* This decentralized blockchain is mainly constructed and maintained by distributed RSUs. Its main responsibility is to complete the decentralized storage of the intrusion detection model aggregated by RSU. In order to incentivize the continuous improvement of model accuracy of RSU, the blockchain of this model adopts a trust incentive mechanism, and the accuracy of the model trained by RSU is used for trust evaluation. The specific consensus adopts the method of POW (i.e., proof of work) plus POA (i.e., proof of accuracy), which encourages RSU to compete to train a high-accuracy model for improving the defensive detection capabilities of the intrusion detection model.

To ensure the security of vehicles' local data, vehicles use their intrusion detection data and network models requested from RSU for training without leaking local data, and then upload the trained model to the specified RSU, without uploading network data, thereby ensuring the privacy and security of their data. As an edge server, RSU can collect trained network edge models of edge vehicles, and train them to aggregate the final network model. RSU has decentralized the central training work by offloading train work to multiple edge vehicles to improve the training efficiency of the network model.

To solve the centralization problem of an RSU, multiple RSUs collect and aggregate their network models at the same time, and implement the final decentralized update and storage of the models through the consensus mechanism and public storage of the blockchain. Blockchain, as a public distributed ledger, is used to implement the model sharing service to solve the inefficiency of repeated model training, thereby improving the efficiency of intrusion detection systems and the security of model sharing. The blockchain system maintained by RSUs not only stores continuously updated machine learning models, but also encourages RSU to train higher accuracy network models with the help of trust evaluation-based incentive mechanism of blockchain.

B. Threat Model

To solve the centralized problem of the model storage of the central aggregation server, this work proposes model sharing and storage based on blockchain. By introducing multiple RSUs to compete and aggregate the final training model, a trust-based incentive mechanism is used to complete the final blockchain consensus. By combining the trust of the RSU and the accuracy of the finally aggregated by the RSU, RSUs are encouraged to compete in training an intrusion detection network model with higher model accuracy.

This work assumes that each edge vehicle uses the global model obtained from the RSU to perform edge training using malicious or random intrusion detection data as the input, and then maliciously uploads low-quality edge model parameters to the RSU. By sending malicious edge parameters to the RSU, edge vehicles can launch poisoning attacks on the RSU, thereby reducing the model accuracy of the RSU aggregation model. When many edge vehicles upload edge model parameters in plain text, adversaries can infer the original training data based on the linkability of users [29] by using the model parameters (e.g., gradients, weights).

If the edge node only uploads plain model information, such as weight parameters or gradient information, the central aggregation server may infer the data owner's data based on the model information, thereby causing privacy leakage. Edge computing wireless communication channels are open and exposed. The adversary can launch speculative attacks on the plaintext parameter information issued by the edge nodes.

IV. Blockchain and Federated Learning for Intrusion Detection

A. Federated Learning Model Broadcasting

Federated learning is used to achieve intrusion detection machine learning for model training. The vehicles V_* are the edge training nodes, and multiple RSUs are served as the aggregators for global model training in its immediate vicinity.

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The edge dataset vector of i th vehicle V_i ($i \in \{1, \ldots, n\}$, n is the number of vehicles scheduled to transmit the pre-trained model) is defined as X_i. The edge weight vector trained on X_i is represented by w_i. When training its network model, each edge node is based on the loss function as follows:

$$L(w, X) = \frac{\sum_{i=1}^{n} L_i\left(w_i, X_i\right)}{n}. \tag{1}$$

Multiple vehicles send a global model request to the RSU, and the RSU dynamically adjusts n according to the different number of vehicle requests received in different periods. n is calculated as follows:

$$n = \gamma C, \tag{2}$$

where γ represents the threshold coefficient, and C is the total number of vehicle requests received by the RSU. γ controls the number of receiving vehicles transmitted and controls the congestion level of the network.

The training process of the model is required to reduce the loss function and eventually converge by

$$w^* = min \quad L(w). \tag{3}$$

RSU uses threshold secret sharing technology to share a secret number s to V_*. RSU first generates a $m-1$ degree polynomial, where $a(0)$ is equal to the secret number s to be shared. Instead, $\alpha_1, \alpha_2, \ldots \alpha_{m-1}$, refers to the random number generated by RSU. Then RSU only needs to send $s_1 = a(1), s_2 = a(2), \ldots, s_n = a(n)$ to $V_1, V_2, \ldots, V_n$ respectively. Any m of $a(1), a(2), \ldots, a(n)$ can solve s together. Any $m-1$ sub-secrets can't get the exact solution of s. The Shamir secret sharing method satisfies addition homomorphism. $a(x)$ is calculated as follows:

$$a(x) = s + \alpha_1 x + \alpha_2 x^2 + \cdots + \alpha_{m-1} x^{m-1} \, mod \, p. \tag{4}$$

To prevent the sub-secrets received by each participant from being attacked by the adversary's continuous observation, dynamic secret sharing can be considered [30]. The validity period of the secret existence is divided into multiple periods, and the fragments of the secret key will be updated at the beginning of each period. This update will not change the shared secret key. Even if the updated fragments are used, the original shared secret key can still be reconstructed. The sub-secret of a participant at time t is recorded as s_i^t. RSU distributes global model weight parameters along with sub-secrets to surrounding vehicles.

Federated learning aggregates model parameters trained by multiple data owners on their devices and then continues training according to the gradient descent algorithm until the loss function converges to finally generate an aggregated global model based on the weight w.

Some representative vehicles can collect edge models and aggregate edge models for pre-training, and then add mask noise before uploading to RSU. To prevent the parameters of the edge model from being transmitted, this model considers adding noise to the parameters to be uploaded, using secret sharing technology, the sub-secret distributed by RSU is added to the model as mask noise, and the aggregator RSU can complete the denoising. When RSU distributes an initial model to a specified vehicle, it also distributes sub-secrets to it, so that it can add mask noise to its edge model parameters.

The RSU collects the edge aggregation model of the edge representative vehicle and performs secret reorganization to remove the mask, and then iteratively trains its own final model based on the aggregated model M. The main notations in this work are listed in Table I.

TABLE I
NOTATIONS

Notation	Description
L	The loss function.
X_i	Edge dataset vector.
w_i	Edge weight vector trained on the specified edge dataset X_i.
$\widetilde{w_i}$	The updated weight with the sub-secret mask after aggregating edge gradients by the edge vehicle.
W_{mask}^i	Edge weights with mask uploaded to RSU.
$\mathbf{g}_i$	The gradient vector.
∇L	The partial derivative function.
$\frac{\partial L_i}{\partial X_i}$	Partial derivative of L_i to X_i.
M	The global aggregated model.
V_*	The vehicle V_*.
s	The secret generated by R.
MSG	A collection of packets from vehicles collected by RSU.

B. Collaborative Edge Learning

A simple multi-layer perceptron neural network is selected to complete the classification task of intrusion detection. Multilayer Perceptron (MLP) is a feedforward artificial neural network model that maps multiple input data sets to a single output data set. The intrusion detection problem can be regarded as a multi-classification problem, so the cross entropy loss function can be used as a loss function as follows:

$$L = -\sum_{r=1}^{Cat} y_k \log\left(\Gamma_k\right), \tag{5}$$

where Cat represents the number of categories, y_k indicates a variable and Γ_k is the predicted probability that the observation sample belongs to category k. The value of y_k could be 0 or 1, which is depending on the category. If the category is the same as the sample, it is 1, otherwise, it is 0.

Each vehicle in an RSU area first requests a global model respectively from the RSU, and then uses the anomaly data collected by it to perform edge learning to train the local model. By using stochastic gradient descent, the weight w_i and gradient of the edge model g_i under the specified data can be calculated and then can be uploaded to its RSU without uploading its data. When each edge vehicle uploads its model updates to its own central RSU, it needs to ensure that its model updates are not intercepted by the adversary, in case the adversary can infer the user's native data based on the model parameters obtained by it.

Each discrete edge vehicle can upload its local model to update to RSU. The problem to be solved by federated learning is to aggregate the global parameter update M without leaking the original data of the edge vehicle. Multiple edge vehicles submit their updated model parameters to the upper-level RSU in a shared manner. Because current inference attacks [31] may infer the user's privacy data based on the model parameters shared by

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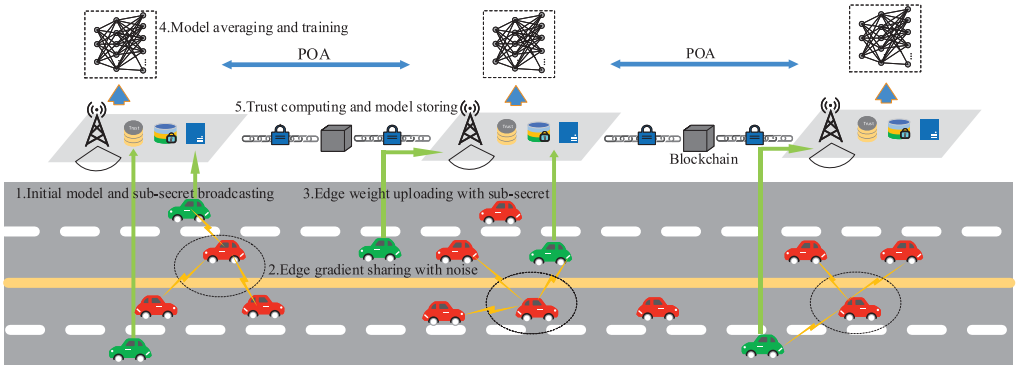


Fig. 3. Edge model sharing and training.

Algorithm 1: Edge Model Averaging and Masking.

Input: global model from the RSU, local datasets of edge vehicles.
Output: edge aggregated model
for each $i \quad 1 \rightarrow n$ **do**
Each vehicle V_i train its model by its dataset and other gradients received from other vehicles;
Add noise into its model:
$\tilde{g}_i^t \leftarrow \frac{1}{n}\sum_{j=1}^{n}(\overline{g}_j^t(x_i) + \text{Laplace}\,(\frac{\Delta f}{\varepsilon}))$;
Share the masked model to other edge vehicle;
end
Receiver aggregated distributed edge model and train the final model with its dataset.
Edge aggregation model adds sub-secret as a mask.
$\widetilde{w}_i^t = w_i^t + s_i^t$
return M;

the user, the uploaded model parameters cannot be uploaded in plaintext.

Fig. 3 shows the whole processing of edge model sharing and training. On the premise of not leaking training data and model parameters, the edge vehicle trains and submits their respective model parameters to RSU.

1) Edge Model Training: Participants are trained on a global model using a specified neural network model (e.g., CNN), avoiding the time and resource consumption caused by training from scratch. To reduce the number of communication times between the edge vehicle and the RSU and reduce the communication overhead, the edge vehicle does not directly upload the gradient information to the RSU after the loss function converges. Instead, the edge vehicles share the gradient information, and then aggregate the edge gradient information to further accelerate the training to obtain a new model weight $\widetilde{w_i}$. The training process of the edge model is mainly divided into two stages. In the first stage, there is no information interaction between edge vehicles. Each vehicle only uploads its model information. Based on the quality of the received edge model, the RSU can evaluate the model quality of the corresponding edge vehicle. After the RSU aggregates the global model, it sends the global model parameter information to other edge vehicles.

2) Edge Model Sharing: In order to improve the model quality of a single edge model, multiple adjacent edge vehicles can enhance their own edge model quality through model sharing and model reception. To achieve the secure sharing of models between vehicles, the edge vehicle shares the edge model after adding differential privacy to the target vehicle. Differential privacy is a method to achieve data protection based on data distortion, protecting the privacy leakage caused by a small change in the data source [32].

Given a similar data set D and D', there is at most one data difference between them. Then given a mapping function $f: D \rightarrow R^d \; f(D) = (z_1, z_2, \ldots, z_d)^T$ which represents the mapping relationship between the data set D and a d-dimensional space. Add Laplace noise $\text{Laplacelace}(\frac{\Delta f}{\varepsilon})$ to the resulting value of the function f to get an output function $A(D)$.

$$A(D) = f(D) + \left(\text{Laplace}_1\left(\frac{\Delta f}{\varepsilon}\right), \text{Laplace}_2\left(\frac{\Delta f}{\varepsilon}\right), \ldots, \text{Laplace}_d\left(\frac{\Delta f}{\varepsilon}\right)\right)^T \tag{6}$$

And Δf is expressed as follows.

$$\Delta f = \max_{D,D'} \left\| f(D) - f\left(D'\right) \right\|_c, \tag{7}$$

Where c generally takes 1 as a norm. Δf satisfies the following formula.

$$P_r[A(D) = O] \leq e^{\varepsilon} * P_r\left[A\left(D'\right) = O\right] \tag{8}$$

As in Algorithm 1, the sender adds its model gradient information to differential privacy, and then the receiver updates its model based on the sender's gradient. The receiver V_i can get multiple gradient information added to the mask noise: $\{\overline{g}_1^t, \overline{g}_2^t, \ldots, \overline{g}_j^t, \ldots, \overline{g}_n^t\}$ $(j \in \{1, \ldots, n\} \& \& j \neq i$, t represents a time slot), based on each gradient, it can be based on its original model, update and train to get convergence weight: $\{\overline{g}_1^t, \overline{g}_2^t, \ldots, \overline{g}_j^t, \ldots, \overline{g}_n^t\} \rightarrow \{w_1^t, w_2^t, \ldots, w_j^t, \ldots, w_n^t\}$. Each vehicle V_i train its model gradient by its dataset and other gradients received from other vehicles as follows:

$$\tilde{g}_i^t \leftarrow \frac{1}{n}\sum_{j=1}^{n}\left(\overline{g}_j^t\left(x_i\right) + \text{Laplace}\left(\frac{\Delta f}{\varepsilon}\right)\right). \tag{9}$$

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Based on the accuracy of the weight of each model, the receiver can abandon the weight model with low model accuracy. Finally, based on the weight information of the sub-model with better accuracy, the receiver performs weighted aggregation and then iteratively trains to get its own final edge model. In this process, the receiver can send warnings to other vehicles and RSUs for the edge model gradient information with lower model accuracy. In this way, other roadside facility vehicles can promptly warn vehicles with lower accuracy to prevent malicious vehicle model sharing. If multiple vehicles have issued a model warning to the model information of the same vehicle, the model information of the shared vehicle will not be accepted.

3) Model Parameters Masking and Uploading: Edge vehicle aggregates the collected multiples of reliable w_j and then averages the model to get w_i. Each vehicle uses a gradient descent algorithm to continuously train on its dataset according to the aggregated edge model until the loss function converges. The participant sends its new model weight $\widetilde{w_i}$ plus the received sub-secret as a mask to the RSU. The masking of the process of the edge model is as Algorithm 1.

Participant V_i who has mastered the shard s at the beginning of the t time period performs the update function to update its shard to s_i^t. V_i first select $m-1$ random numbers $\alpha_{i1}, \alpha_{i2}, \ldots, \alpha_{i(m-1)} \in {}_R Z_p$ to define polynomial as follows:

$$\alpha_i(x) = \alpha_{i1}x + \alpha_{i2}x^2 + \cdots + \alpha_{i(m-1)}x^{m-1}, \tag{10}$$

V_i then computes $u_{ij} = \alpha_j$ and sends u_{ij} to other model sharer V_j. After receiving u_{ji} from all other participants, the vehicle updates its fragments according to

$$s_i^{(t)} = s_i^{(t-1)} + (u_{1i} + u_{2i} + \cdots + u_{ni}), \quad j \in [1, \ldots, n] \&\& j \neq i. \tag{11}$$

After updating the sub-secret, V_i destroys all other quantities.

4) Aggregation and Storage of Shared Models by RSUs: After receiving at least m gradient update data, the RSU as a receiver can aggregate them together. The aggregated results are accompanied by the aggregated secret s as a mask, in addition to the aggregated gradient update. RSU subtracts the secret value s to offset the secret value as a mask and can obtain a pure aggregation gradient. M is the final model to be aggregated. U represents the subscript set of edge nodes to which all received gradients belong. The process of removing the mask is calculated as follows:

$$\begin{aligned} M &= \sum_{u\in U} \widetilde{w}_u + \sum_{u\in U} \lambda_u s_u^{(t)} - s \\ &= \sum_{u\in U} \widetilde{w}_u + \sum_{u\in U} \lambda_u \left(s_u^{(t-1)} + \sum_{v\in U\setminus\{u\}} \alpha_v(u) \right) - s \\ &= \sum_{u\in U} \widetilde{w}_u + \sum_{u\in U} \lambda_u s_u^{(t-1)} + \sum_{v\in U\setminus\{u\}} \sum_{u\in U} \lambda_u \alpha_v(u) - s \\ &= \sum_{u\in U} \widetilde{w}_u + s + \sum_{u\in U} \alpha_v(0) - s \\ &= \sum_{u\in U} \widetilde{w}_u. \end{aligned} \tag{12}$$

Algorithm 2: Global Model Averaging and Aggregation.

Input: $MSG = \{MSG_u | u \in U\}$, Model: M, Train dataset: X
Output:
$Accu = [Accu_1, Accu_2, \ldots Accu_i, \ldots Accu_{m+1}]$
Initialize Accu = [];
for each $u \in U$ **do**
Calculate $\hat{MSG}_u = MSG - MSG_u$;
Aggregation Calculate $w = \sum_{u\in U} w_u$;
Train $M(w, X)$;
Calculate $Accu_u$;
Accu += $Accu_u$;
end
return $M_u, Accu_u$;

The model averaging and aggregation algorithm is described as Algorithm 2. Under a (m, n) threshold secret sharing technology, this model assumes that an RSU can receive the edge model parameter information of at least $m+1$ vehicles, then any m of the vehicles can reconstruct the secret s, and the secret mask is removed to obtain the aggregation weight. For a collection of information $MSG = \{MSG_u | u \in U\}$ made up of packets sent by each vehicle $MSG_u = (W^u_{mask}, VID_u)$, the RSU can take any t packets of them to build a new data set $\hat{MSG}_u = MSG - MSG_u$ which does not contain MSG_u for constructing aggregation weights. The RSU can track the quality of the each MSG_u model passed by some vehicle according to the accuracy of the averaged model trained by $(m+1)$ $\hat{MSG}_u$. Based on this collaborative monitoring average model, each masked weight packet uploaded by each vehicle is traceable.

A complete intrusion detection system should be able to resist both external and internal attacks. By establishing trust management for connected vehicles, internal attacks between connected vehicles can be effectively resolved. There are two types of vehicles that do not participate in cooperation. One is a malicious vehicle, and the other is a selfish vehicle [33]. Through model quality evaluation, malicious vehicles can be effectively identified. When multiple vehicles share gradient information, considering the internal attack of fraudulent sharing of information between vehicles, a trust system should be established between the vehicles. Vehicles in a certain RSU area can form a cluster in which interconnected vehicles can build a trusted network based on blockchain for data sharing.

C. Trust Computing and Blockchain Consensus

When the edge vehicle requested a global model from the RSU, the RSU was able to distribute the global model stored in the blockchain to the edge vehicle. The blockchain relies on a peer-to-peer network structure, where each RSU node has a copy of the global machine model. Similar to a distributed database, a blockchain-based data storage and sharing model can effectively avoid the risks of a single point of downtime and a single point of attack. As an open ledger, the blockchain can effectively guarantee the integrity and authenticity of the original global model. The blockchain-based incentive and punishment

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mechanism will prompt participants to upload correct gradient information. The transaction behavior of each block is traceable.

The upper-level blockchain model as model-chain is built between multiple RSUs. The model-chain mainly stores the global model of intrusion detection at different times. At a new point in time, each RSU can aggregate and train its global model to conduct election absenteeism with higher accuracy. After each RSU receives model parameters from other RSUs, it performs test verification using shared intrusion detection data to achieve blockchain consensus.

Different RSUs aggregate their edge model parameters that they collect from the edge vehicles and then iteratively train to obtain their aggregate models. The aggregation work of each RSU at different time points can be considered as a training transaction. The original intrusion detection data is shared among multiple RSUs on the same road segment. After the distributed RSU aggregates its model, it shares its machine learning model, and model accuracy rate tested on public datasets in the peer-to-peer mode. Each RSU can train and verify these models and model accuracy rates received from other RSUs to check the accuracy and authenticity of each RSU's model. Multiple RSUs jointly maintain a blockchain in which the consensus mechanism adopts the POA. The quality of the training model is used as the blockchain consensus. The RSU with a higher accuracy model can write the aggregation model into the blockchain. The incentive mechanism can stimulate RSU participants to actively participate in collaborative training, thereby training the final intrusion detection model.

Multiple RSUs send information to each other, and the broadcast information content of the sender mainly includes global model parameters, model accuracy, sender's identity information, and data signature. To encourage multiple RSUs to participate in the training of highly accurate aggregation models, this model implements trust evaluation and trust management for RSUs based on the accuracy of the training models. When an RSU contributes to a higher accuracy model, its trust value will increase accordingly. Under this trust-based incentive mechanism, multiple RSUs can compete in training a global model with high accuracy.

1) Calculation of the Offset of Trust Computing for RSUs: After all RSUs aggregate their own model M_b ($b \in B$, B is the subscript set of all RSUs), each RSU broadcasts its model M_l ($l \in B \&\& l \neq b$) to other RSUs, and also receives the models broadcast by other RSUs. Each RSU M_b first performs training verification based on the other received model collections $\{M_l\}$ to obtain the accuracy set $\{A_l\}$. Calculate the accuracy deviation value β_b^t between the RSU and other RSUs based on the accuracy set. β_b^t is accuracy deviation value which is calculated as follows:

$$\beta_b^t = (1 + \frac{\sum\limits_{l \in B \setminus b} (A_b - A_l)}{|B| - 1}). \tag{13}$$

The trust offset index indicates the quality superiority of the RSU relative to other RSUs. According to the calculated offset coefficient, the current RSU can perform trust update calculation based on the historical trust value as follows:

$$S_{R_b}^t = \frac{1}{1 + e^{-\alpha \cdot (S_{R_b}^{t-1} \cdot \beta_b^t)}}, \tag{14}$$

where $S_{R_b}^t$ is the trust value of R_b at the previous time slot t, and A_b is the model accuracy of RSU R_b. The trust value is between 0 and 1. The greater the deviation index, the faster trust value grows. After an RSU with the highest model accuracy is verified by other RSUs, the RSU will package the model information that finally obtained the consistency verification and these updated trust values of all RSU participants into a new block.

2) Incentive Mechanism Based on Trust Reward: Throughout the incentive process, the accuracy of the model is linked to trust evaluation. When the accuracy of an RSU is higher than the accuracy of most RSUs, then its trust value will increase accordingly. On the contrary, it will decrease. The incentive mechanism of the blockchain will prompt RSU to compete in training models with higher accuracy. Based on the trust evaluation of RSUs, after a certain R_b receives the model of R_l, it first queries the trust value of R_l from the blockchain before verifying the accuracy of the model of R_l. If it is less than a certain threshold value, even if the machine learning model published by R_l has a high accuracy rate, the model will also be considered untrustworthy.

After calculating the trust value, the RSU with the highest trust value will successfully run as a winner. In order to enable RSUs to honestly and autonomously participate in the sharing and increasing of accurate models, after each time slot of trust update, the RSU with a higher accuracy rate of the contribution model is given a trust reward, and the magnitude of the reward is mainly based on the proportion of the rate the model offset, if the current model's offset rate β is greater than a certain threshold ξ, then it will reward it with increased trust, otherwise, it will be punished with reduced trust. The calculation process is as follows:

$$S_{R_l}^t = \begin{cases} S_{R_l}^t - \frac{\eta_l e^{A_l}}{\sum_{r=1}^{z} \eta_r e^{A_r}}, & 0 < \beta < 1, \\ S_{R_l}^t + \frac{\eta_l e^{A_l}}{\sum_{r=1}^{z} \eta_r e^{A_r}}, & 1 <= \beta. \end{cases} \tag{15}$$

The value of ξ_t is the maximum value of the model offset rate at the last moment

$$\xi_t = \max(\beta_b^t | b \in B). \tag{16}$$

The total cumulative trust value of RSU can be denoted as S_{R_b} and the final trust value evaluation of the system is as follows:

$$max \left(S_{R_b} = \left[\sum_{b \in B} S_{R_b}^t \left(\alpha, \beta_b^t \right) \right] \right). \tag{17}$$

Since the consensus of this blockchain mainly uses POA, according to the hash calculation principle of the blockchain, the number of zeros on the left in the binary hash value of the blockchain increases according to the decrease of the number of zc. Therefore, the lower the difficulty of mining nodes with a higher trust value, the RSU with a higher trust value is more likely to find a suitable $nounce$ to make mining successful

$$Z = \lfloor \left(e^{-\left(\eta \cdot S_{R_i} + \mu \right)} \right) \rfloor. \tag{18}$$

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D. Security Analysis

This section discusses the security analysis of the proposed approach, where internal and external attacks can be avoided. The IDS continuously collects edge model parameters updated by edge vehicles. With multiple parameter uploads of different vehicles, vehicles and RSUs jointly maintain a dynamically updated intrusion detection system.

- *Information speculation from edge model:* The adversary infers the raw data of the edge vehicle based on the weight and gradient information of the intercepted edge model. To prevent the upper-level aggregation server from inferring the native data information of the edge vehicle based on the uploaded gradient information, each vehicle adds noise to the gradient information uploaded by itself. To make the system more lightweight, when the vehicle uploads the gradient, instead of encrypting the gradient information, it uses the secret-sharing method based on the (m, n) threshold with the sub-secret received from RSU. The vehicle uses the sub-secret as a mask to mix the gradient information and then send the mixed mask data to the RSU. After receiving the information of at least m vehicles, RSU can reconstruct the original secrets and then remove the unique secrets, to obtain the noise-free aggregate model parameters, but the RSU does not get the pure edge model parameters uploaded from any edge vehicle in the entire process, so it is impossible to infer the raw data information of the edge vehicle. In a cluster, for two data D and D' of an edge vehicle, the probability densities are numerator and denominator respectively. According to the definition of differential privacy sensitivity, it can be derived from the triangle inequality that for a certain query set S, its single query r satisfies ε differential privacy.
- *Robustness analysis:* For an edge community composed of edge vehicles in a certain RSU area, if one of the vehicles is attacked, the training and aggregation of the global model of the community will be affected. Considering the robustness of the system, it should still be able to work when some of these vehicles are offline. Considering the unsafeness and unreliability of some vehicles, RSU does not directly aggregate the model parameters of all the edge vehicles it received when it aggregates. Based on the (m, n) threshold mechanism, this RSU can select m vehicles' information uploaded from the n vehicles received for aggregation. This system assumes that at least m vehicles' model data can be uploaded. In this system, m can be dynamically adjusted according to the condition of the communication link or traffic conditions in a certain period.
- *Poisoning attack:* If an edge vehicle uploads malicious and poisoning updates, the training of the global model may be polluted. To avoid some edge vehicles from deliberately uploading poor-quality edge model parameters, this system designed a data tracking and model selection mechanism for RSU. Suppose the edge model $MSG_u = (W^u_{mask}, VID_u)$ uploaded by the vehicle V_u has certain toxicity. In order to track down the poisonous model MSG_u, the system considers the aggregation model collection $S_1 = \{\hat{MSG}_{untake}\}$ when MSG_u does not participate, and MSG_u participates in which the aggregation model set of the situation is $S_2 = \{\hat{MSG}_{take}\}$. For the above two aggregation model sets, the statistical training accuracy set is $A_1 = \{Accu\}$ and $A_2 = \{Accu\}$ respectively. According to the combination principle, the sizes of the two sets are C^m_{n-1} respectively. Pick the larger value in A_1 and A_2 respectively and calculate the difference. If the difference is less than a certain threshold m_α, then the system considers that the model $MSG_u = (W^u_{mask}, VID_u)$ has a poison attack. Under the (m, n) threshold mechanism, RSU can aggregate edge model parameters of different t groups. Then by comparing the accuracy of the aggregated models of different groups, for some aggregate models with lower accuracy, RSU can infer which model parameters uploaded by the edge vehicle caused the low accuracy of the aggregate model. In the end, after the aggregation, the RSU can select a certain aggregation model with higher accuracy as its aggregation result. According to this tracking and selection mechanism, the system can effectively avoid the poisoning attack on the aggregation process caused by the intentional upload of poor quality models by the edge nodes.

TABLE II
EXPERIMENTAL PARAMETERS

Paramaters	Values
Hash Function	SHA-256
η	0.1
μ	1
α	0.1
γ	0.8
the length of secret s	256 bits
dataset	KDDCup99

V. IMPLEMENTATION AND PERFORMANCE ANALYSIS

The third-party libraries Pytorch for deep learning and Syft for federated learning is used for implementation [34], and the system model is simulated based on Python programs on a Mac Os Catalina workstation with the Quad-Core Intel Core i5, 64-bits, 2.3 GHz PC.

The data set used in this experiment is KDDCup99 [35]. This data set is 9 weeks of network connection data collected from a simulated US Air Force LAN, divided into labeled training data and unlabeled test data. The test data and the training data have different probability distributions, and the test data contains some types of attacks that do not appear in the training data, which makes intrusion detection more realistic. Each connection record in the KDDCup99 training data set contains 41 fixed feature attributes and 1 classification identifier. The identifier is used to indicate that the record is normal or a specific type of attack. Among the 41 fixed feature attributes, 9 feature attributes are discrete, and the others are continuous. When the size of the training data is 4000, the size of the test data is 4000, and the epoch is 40.

The main parameters used in the experiment are listed in Table II. The experimental parameters selected here are mainly used to observe the subsequent experimental results more conveniently. For example, α is used to control the variation range of

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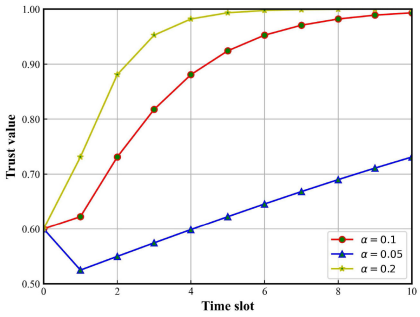


Fig. 4. The trust value changes in different α.

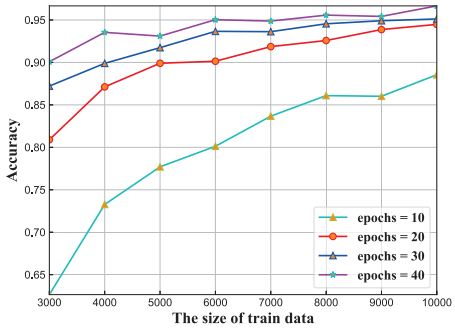


Fig. 5. The accuracy rate changes in relation to data size.

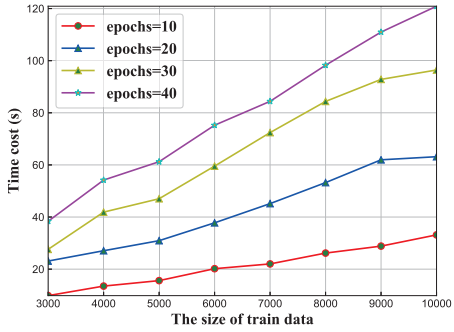


Fig. 6. Running time in various train data sizes.

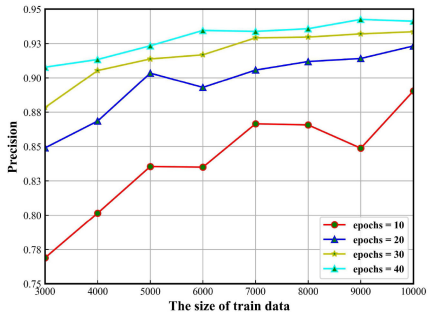


Fig. 7. The precision rate changes in relation to data size.

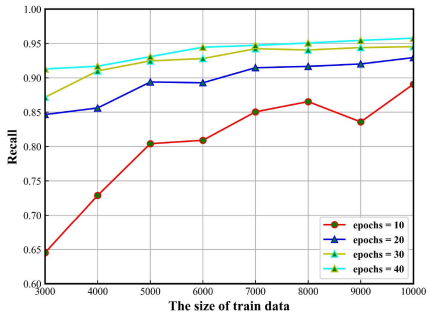


Fig. 8. The recall rate changes in relation to data size.

the trust value in different time periods. We initialized the trust value of all vehicles to 0.6 and simulated how the trust value of well-behaved vehicles changed with different alpha values. The experimental results showed that when α was set to 0.05, trust values went down at first and then gradually went up, which is not consistent with the vehicle behavior. When α was set to 0.2, the trust value of the vehicle changes greatly, which is not conducive to observing the influence of the trust value on the accuracy of its model. In order to prevent the trust value from being updated too quickly in a certain time period, we set a smaller value of 0.1 to facilitate us to observe the influence of model accuracy on the trust value in a long time period. The parameters η and μ are mainly related to the difficulty of mining in the consensus algorithm. We set η to 0.1 and μ to 1 so that POW and POA algorithms have similar mining difficulty at the beginning, which makes it convenient to compare them.

The experiment tested the accuracy of the model under different data sets and different epochs. The epoch represents the training times of the model. Fig. 5 shows that under the same epoch, the accuracy rate increases with the increase of the data set. Under the same size of the dataset, the accuracy rate will also increase with the increase of epoch, and when the epoch is in the range of 30 to 40, the increase is not obvious, which shows that when the epoch is close to 30, the model basically reaches convergence.

Fig. 6 shows that the training time of the model changes under different data set sizes and different epochs. When the amount of data is the same, the training time of the model increases as the epoch of the model changes. From the growth rate of the curve, when the epoch is between 30–40, the increase in time is lower than that of the epoch between 20–30. The convergence rate of the model is faster when the epoch is between 30–40.

Fig. 7 and Fig. 8 show that under the same epoch, the precision rate and the recall rate increase with the increase of the data set. The experimental results showed that the precision and recall of the proposed algorithm could reach a high value under the condition of sufficient training data sets and epochs. When the training data set reached 10 000, and the epochs were equal to 40, the precision was close to 94 percent, which indicated that when the prediction result of this model is an attack, its reliability is high. At the same time, the recall reached 95, indicating that most of the attack samples can be detected.

Fig. 9 shows that changes in the accuracy of the RSU aggregation model when different numbers of edge vehicles collaborate in the training process of federated learning. It can be seen from the figure that under different numbers of edge nodes, the changing trend of the model accuracy rate is consistent. Because under federated learning, each edge vehicle has equal status as an edge node. After uploading its model parameters to RSU,

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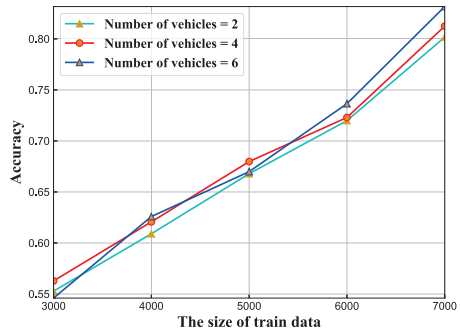


Fig. 9. Accuracy in various number of vehicles.

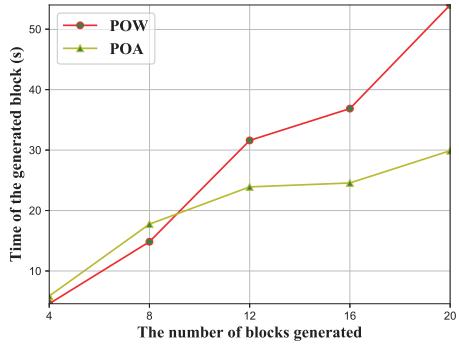


Fig. 10. The consensus algorithms on block generation time.

RSU performs average arithmetic aggregation on the model parameters uploaded by multiple uploaders. As a result, when different numbers of edge vehicles participate in the training, the changing trend of the accuracy rate of the aggregate model is the same.

A public blockchain network is built by using go language. The implementation focused on the public Ethereum blockchain consensus mechanism, and made improvements on the basis of the traditional POW consensus mechanisms, and evaluated the time loss under different consensus mechanisms. The block generation time under different consensus algorithms was simulated when the number of blocks was generated by using the Go language. By using the trust value as the equity, the relationship between the hash value and the trust value is established, thereby simulating a node with a high degree of trust, making it easier to successfully mine.

Fig. 10 shows that the impact of different consensus algorithms on block generation time. It can be seen from the figure that in the early stage, there is not much difference between the block generation time of POA and POW consensus. After a certain amount of blocks are generated, the time of POA is obviously lower than that of POW, which is mainly due to the trust-driven nature of the POA. When distributed computing nodes do not accumulate a certain amount of trust, they still mine according to the POW method. After a certain amount of trust is accumulated, the difficulty of mining will decrease due to the increase in trust, so the time of mining the block time is reduced accordingly. Through this experiment, it can be proved that the POA algorithm proposed in this paper can effectively improve the efficiency of POW algorithm to generate blocks.

VI. Conclusion

This work proposed a blockchain and federated learning for collaborative intrusion detection for vehicular edge computing networks. The proposed scheme achieved the final higher accuracy model aggregation and sharing through multi-party aggregation training. To improve the quality of the model, this work proposed a blockchain incentive mechanism based on trust and the quality of training model, which motivates distributed RSUs to complete in training high-accuracy model and secure storage and sharing.

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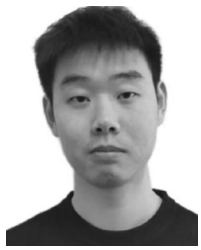
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